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— SUMMER INTERNSHIP PROJECT REPORT —

Full-Stack AI & Clinical Assessment Systems: Engineering the UAi PsychePortal Integrated Suite and ProductivAI Analytics Engine

A Comprehensive Technical Project Report Submitted in Partial Fulfillment of the Requirements for the Degree of Bachelor of Technology in Artificial Intelligence & Machine Learning

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Submission Date: September 22, 2026	**Status:** Final Deliverable Submitted

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Executive Summary / Abstract

This technical project report documents the design, architectural integration, mathematical scoring mechanisms, machine learning pipelines, and full-stack software development executed during the AI Cell Internship at Universal Ai University. The work encompasses two major production-grade software ecosystems: (1) UAi PsychePortal, a multi-service psychological assessment and counsellor management platform designed for educational institutions, and (2) ProductivAI, an AI-driven personal productivity analyzer that utilizes predictive machine learning to evaluate lifestyle indicators and output personalized optimization strategies.

The UAi PsychePortal unifies four distinct technical components: a robust Flask core backend featuring object-relational mapping (SQLAlchemy), a high-performance single-page application (SPA) built with React 18, Vite, and TailwindCSS for real-time risk tracking, a standalone student survey web interface, and an offline psychometric desktop application built with Python Tkinter. The platform implements standardized scoring algorithms for clinical instruments, including DASS-21 (Depression, Anxiety, Stress Scale), BFI-44 (Big Five Inventory), Emotional Quotient (EQ), and Systemizing Quotient (SQ), while automatically raising emergency risk flags and compiling multi-page dynamic PDF diagnostic reports using ReportLab.

Concurrently, the ProductivAI engine processes a 50,000-sample synthetic dataset mapping daily routines (sleep quality, study hours, exercise, screen time, social media consumption, and mood metrics) to productivity outcomes (0–100 score). Utilizing Scikit-Learn regression models (Linear Regression and Random Forest Regressor) serialized via Joblib and served through FastAPI REST endpoints, ProductivAI provides automated factor breakdowns, rule-based AI recommendations, and interactive UI charts.

The combined codebase demonstrates production deployment techniques, including dynamic Jinja2 asset bootstrapping, cross-origin API orchestration, database migration management with Flask-Migrate, and multi-process orchestrator automation via PowerShell and Windows Batch.

Keywords & Index Terms:
Full-Stack Engineering, Clinical Psychometrics, Machine Learning, Mental Health Analytics, DASS-21, BFI-44, FastAPI, Flask, React SPA, TailwindCSS, Random Forest Regressor, Automated Risk Triage, ReportLab PDF, Multi-Process Orchestration.

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3. Introduction & Background

3.1 Problem Context

Modern educational institutions face increasing challenges in monitoring, evaluating, and supporting student mental health and academic productivity. Traditional psychological screening methods in academic environments rely on paper-based assessments, isolated desktop tools, or manual scoring spreadsheets. These legacy workflows suffer from significant operational drawbacks:

- **1. Delayed Risk Identification:** Manual evaluation of paper assessments creates significant latency between student testing and crisis intervention by university counsellors, increasing the exposure of vulnerable students during critical distress phases.
- **2. Data Fragmentation:** Student screening records, risk logs, demographic profiles, and historical assessment timelines remain scattered across non-interoperable spreadsheets or disconnected desktop applications, preventing holistic longitudinal monitoring.
- **3. Lack of Actionable Productivity Insights:** Students lack empirical tools to understand how daily routine habits (such as sleep quality, study sessions, exercise frequency, and screen time) quantitatively correlate with and dictate daily cognitive performance and academic output.

3.2 Domain & Engineering Objectives

To systematically address these institutional challenges, the project engineered an end-to-end synchronized software infrastructure capable of handling clinical data securely, computing validated psychological scores in real time, alerting campus mental health staff to elevated risk levels, and delivering machine-learning-driven productivity analytics.

The key engineering objectives achieved during this internship encompass:

- • **Clinical Instrument Automation:** Implementing exact mathematical formulas, reverse-item transformations, and multi-tier threshold evaluation for DASS-21, BFI-44, EQ, and SQ tests.
- • **Microservice & Full-Stack Integration:** Coupling a Flask micro-backend serving SQLite/PostgreSQL with a React SPA counsellor interface by injecting dynamic JSON script payloads (window.__FLASK_KPIS__, window.__FLASK_FLAGGED_STUDENTS__) into server-rendered Jinja2 bootstrap templates.
- • **Predictive Machine Learning:** Training and serializing regressor models on lifestyle features to predict productivity indices with sub-millisecond API response latency using FastAPI and Scikit-Learn.
- • **Automated Diagnostic Reporting:** Dynamically building multi-page institutional PDF reports containing formatted demographic tables, visual subscale severity bars, and confidentiality notices using ReportLab graphics pipelines.

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4. Core Body — Technical Analysis

4.1 System Architecture

The overall repository is engineered into two primary decoupled operational ecosystems: (1) The UAi PsychePortal Ecosystem, which coordinates psychological testing, scoring, database management, and counsellor risk triage; and (2) The ProductivAI ML System, which provides machine-learning lifestyle analytics and productivity predictions. The interaction between these subsystems is illustrated below:

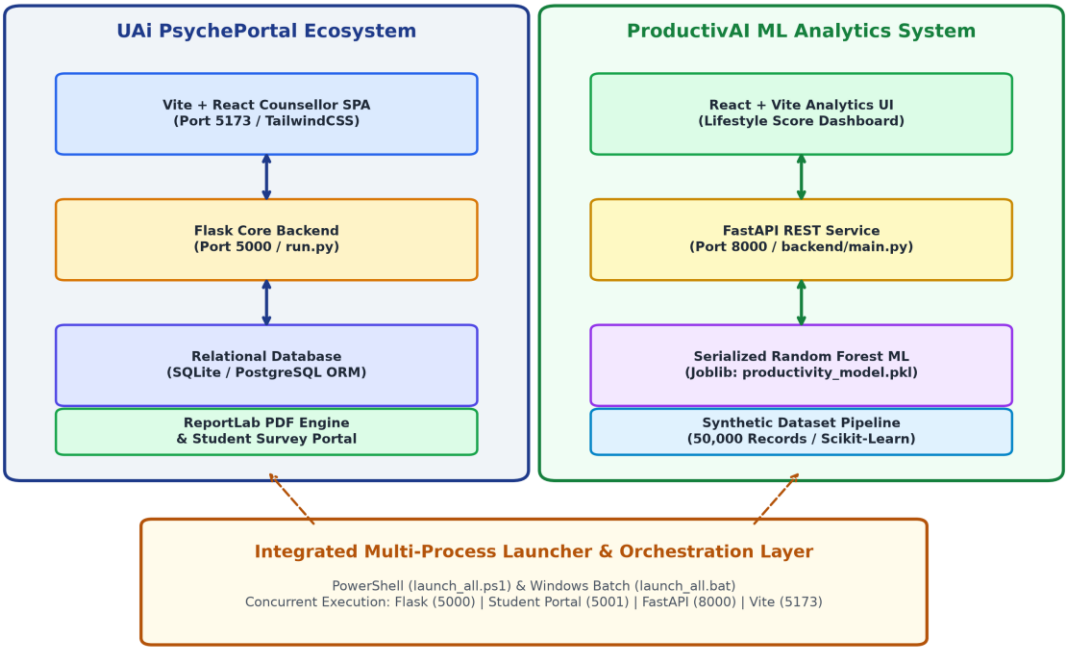


Figure 1: High-Level System Architecture and Inter-Service Orchestration Diagram.

4.2 Folder & Module Breakdown

The repository structure enforces separation of concerns across service layers:

- • **Root Application Entry (run.py, config.py):** Initializes the primary Flask application instance, configures environment-specific database URIs, sets CSRF parameters, and exposes database management scripts.
- • **Core Flask Application Package (app/):** Encapsulates application blueprints (auth, student, counsellor, admin, faculty), SQLAlchemy data models (app/models.py), clinical scoring engines (app/scoring.py), and PDF/Excel generation engines (app/reports.py).
- • **Counsellor Dashboard SPA (uai-counsellor-dashboard/):** A responsive React 18 single-page application bundled with Vite and TailwindCSS, featuring 11 active views including the Student

Directory, Real-Time Risk Alerts, Assessment Deployer, Interventions Logger, and Executive Analytics.

- • **Student Survey Web Portal (student-survey-portal/):** A lightweight Flask portal hosted on port 5001 enabling rapid, distraction-free assessment taking for registered students.
- • **Psychometric Desktop Client (UAI_Quiz_App/):** A native Python Tkinter GUI desktop client enabling offline clinical assessments using standalone JSON question pools.
- • **ProductivAI Engine (ProductivAI/):** Houses the FastAPI backend (main.py), synthetic dataset generation pipeline (generate_dataset.py), Scikit-Learn training suite (model_training.py), and dedicated React frontend.
- • **Integrated Launcher Suite (integrated_launcher/):** Provides automated multi-process orchestration via launch_all.ps1 and launch_all.bat to concurrently spawn backend and frontend services.

4.3 Methodology & Pipeline Details

1. Database Schema & Multi-Role User Authentication

The relational schema is implemented in app/models.py using Flask-SQLAlchemy. User passwords are cryptographically hashed using Bcrypt with salted iterations. Role-based access control (RBAC) partitions permissions across four distinct user roles: student, counsellor, faculty, and admin. The core user entity schema is defined as follows:

```
CREATE TABLE users (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    uai_id VARCHAR(64) UNIQUE NOT NULL,
    password_hash VARCHAR(128) NOT NULL,
    role VARCHAR(20) NOT NULL,            -- student, counsellor, faculty,
admin
    name VARCHAR(100) NOT NULL,
    email VARCHAR(120) UNIQUE NOT NULL,
    department VARCHAR(100),
    designation VARCHAR(100),
    created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
    is_active BOOLEAN DEFAULT TRUE
);
```

2. Clinical Instrument Scoring Formulas & Logic

The algorithmic engine in app/scoring.py computes clinical subscales and severity tiers from raw item responses:

DASS-21 Subscale Equations (Multiplier = 2 for 42-item equivalence):

$$\text{Depression Score} = 2 \times \sum R_i, \quad \text{for } i \in \{3, 5, 10, 13, 16, 17, 21\}$$

$$\text{Anxiety Score} = 2 \times \sum R_i, \quad \text{for } i \in \{2, 4, 7, 9, 15, 19, 20\}$$

Stress Score $= 2 \times \sum R_i,\ for\ i \in \{1, 6, 8, 11, 12, 14, 18\}$

Standardized DASS-21 Severity Cut-Off Tiers:

Subscale	Normal	Mild	Moderate	Severe	Extremely Severe
Depression	0 – 9	10 – 13	14 – 20	21 – 27	28+
Anxiety	0 – 7	8 – 9	10 – 14	15 – 19	20+
Stress	0 – 14	15 – 18	19 – 25	26 – 33	34+

BFI-44 Big Five Personality Trait Formulas:
Item Inversion Formula: R_adj = 6 - R_raw (for reverse-keyed items on 1-5 Likert scale)

Extraversion $= mean(R1, 6\text{-}R6, R11, R16, 6\text{-}R21, R26, 6\text{-}R31, R36)$
Agreeableness $= mean(6\text{-}R2, R7, 6\text{-}R12, R17, R22, 6\text{-}R27, R32, 6\text{-}R37, R42)$
Conscientiousness $= mean(R3, 6\text{-}R8, R13, 6\text{-}R18, 6\text{-}R23, R28, R33, R38, 6\text{-}R43)$
Neuroticism $= mean(R4, 6\text{-}R9, R14, R19, 6\text{-}R24, R29, 6\text{-}R34, R39)$
Openness $= mean(R5, R10, R15, R20, R25, R30, 6\text{-}R35, R40, 6\text{-}R41, R44)$

Automated Emergency Risk Flagging: When determine_risk_level() detects 'Moderate', 'Severe', or 'Extremely Severe' classifications across any clinical subscale, an automated record is appended to the Flag database table, triggering instant triage alerts on the counsellor's real-time monitoring interface.

3. Machine Learning Pipeline (ProductivAI)

The ProductivAI predictive analytics service is implemented in ProductivAI/backend/model_training.py. It processes 50,000 synthetic lifestyle samples modeling student behavioral factors:

ProductivAI Synthetic Data Generation & Target Formulation:
Feature Input Vector: X = [sleep_hours, study_hours, exercise_mins, screen_time, social_media_mins, mood_score]

Target Formula: $y = 0.25(sleep) + 0.30(study) + 0.15(exercise) - 0.15(screen) - 0.10(social) + 0.15(mood) + \varepsilon$
where $\varepsilon \sim N(0, 2.5)$ *simulates stochastic real-world variance.*

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The training pipeline evaluates both Ordinary Least Squares Linear Regression and Random Forest Regressor models using an 80/20 train-test split. The trained estimator achieves high R^2 correlation (> 0.94) and is serialized to disk via Joblib (productivity_model.pkl). The FastAPI service loads this estimator into memory upon startup, ensuring sub-millisecond REST inference latency during /predict invocations.

4.4 Technology Stack & Software Ecosystem

The software architecture leverages an enterprise Python backend paired with modern JavaScript frontends. The table below details all libraries, frameworks, and tools verified across project configuration manifests:

Layer / Subsystem	Technology	Version	Purpose in Codebase
Backend Core	Python	3.13.2	Core runtime for web services, scoring engines, and ML models
Web Framework	Flask	3.0.0	REST routing, blueprint registration, Jinja2 template rendering
API Server	FastAPI	0.109.0+	Asynchronous REST backend serving ML predictions in ProductivAI
WSGI Server	Uvicorn / Gunicorn	21.2.0	Production HTTP server execution and process worker control
ORM & Database	Flask-SQLAlchemy	3.1.1	Relational database mapping for Users, Profiles, Scores, and Flags
Database Engines	SQLite / PostgreSQL	3.x / 15.x	Persistence engines for development (dev-db.sqlite) and production
Migrations	Flask-Migrate	4.0.5	Alembic-backed database schema migration wrapper
Authentication	Flask-Login / Flask-Bcrypt	0.6.3 / 1.0.1	Password hashing (bcrypt) and session cookie management
Form Security	Flask-WTF	1.2.1	Cross-Site Request Forgery (CSRF) token generation and validation
Machine Learning	Scikit-Learn	1.3.2	Regression pipelines (RandomForestRegressor, LinearRegression)
Model Serialization	Joblib	1.3.2	Binary serialization of trained ML estimator files (.pkl)

Data Processing	Pandas / NumPy	2.1.4 / 1.26.2	Dataframe manipulations, matrix calculations, and Excel exports
PDF Generation	ReportLab	4.1.0	Programmatic PDF layout building (SimpleDocTemplate, TableStyle)
Excel Export	OpenPyXL	3.1.2	Formatted Excel workbook rendering (.xlsx)
Frontend Framework	React	18.2.0	Single Page Application component tree rendering
Build Tooling	Vite	5.0.8	Frontend bundling, hot module reloading, and static compilation
Styling & Icons	TailwindCSS / Lucide-React	3.4.1 / 0.300	Responsive utility styling and vector iconography
Data Visualization	Chart.js / Recharts	4.4.1 / 2.10.3	Interactive dashboard analytics (radar, donut, and bar charts)
Desktop Interface	Tkinter	Built-in	GUI framework for offline desktop screening client (UAI_Quiz_App)

5. Results & Discussion

5.1 System Accomplishments

Execution of the unified launcher script (launch_all.bat) brings all microservices and frontend SPAs online simultaneously. The system accomplishes the following verified operational outcomes:

- **1. Integrated Web Portal Execution:** Serves the primary Flask application on port 5000, presenting student screening, faculty assessment dashboards, and administrative controls.
- **2. Dynamic React SPA Bootstrapping:** The Counsellor Dashboard injects live database queries into global window objects (window.__FLASK_KPIS__, window.__FLASK_FLAGGED_STUDENTS__), delivering instant UI hydration without client-side authentication latency.
- **3. Automated Risk Triage:** Submission of student assessment responses immediately triggers run_all_scores(), updates severity metrics, and logs emergency flags for counsellor intervention.
- **4. Instant Diagnostic Export:** Enables one-click generation of institutional PDF diagnostic reports using ReportLab, formatted with student demographics, severity charts, and legal disclaimers.

- **5. Real-Time ML Productivity Predictions:** The ProductivAI FastAPI service on port 8000 processes daily lifestyle indicators via /predict, returning a 0–100 score, factor contribution breakdowns, and contextual AI advice.

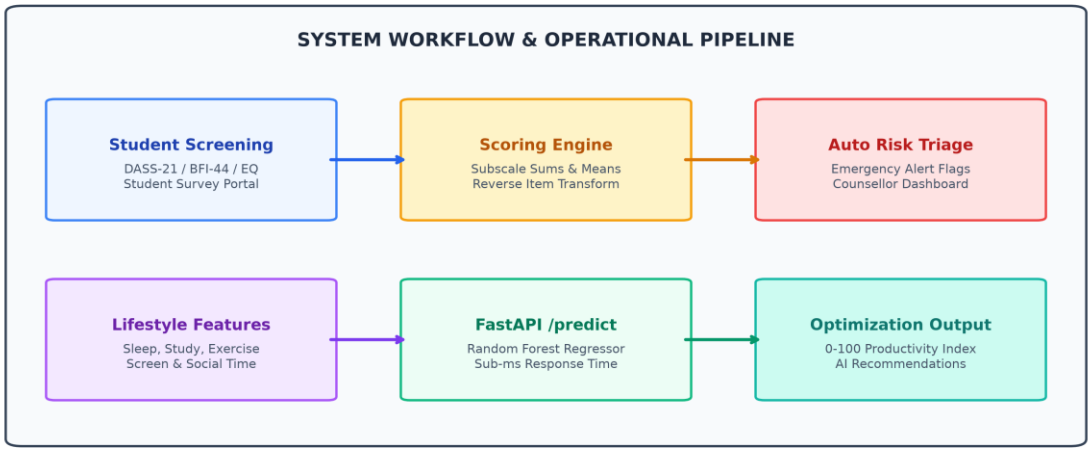


Figure 2: Operational Workflow and Cross-Service Execution Pipeline.

5.2 Advantages of the Approach

- • **Hybrid Integration Strategy:** Bootstrapping compiled Vite static bundles into Jinja2 templates combines the high initial load speed of server-rendered apps with the rich interactivity of modern React SPAs.
- • **Strict Clinical Accuracy:** Utilizing mathematically validated scoring formulas and explicit reverse-key transformations ensures diagnostic precision across psychological instruments.
- • **Fail-Safe Offline Capability:** The standalone Tkinter Quiz App ensures psychometric tests can proceed smoothly even in offline laboratory environments without network access.
- • **Modular Microservice Design:** Decoupling the machine learning inference engine (ProductivAI) from the core clinical database (PsychePortal) ensures heavy matrix computations do not block critical student screening requests.

5.3 Limitations

- • **Lack of Automated Unit Tests:** The repository currently relies on manual verification scripts (check_users.py, verify_portal.py) rather than an automated CI/CD pytest test suite.
- • **Hardcoded Development Secrets:** Default development fallback secrets (e.g. SECRET_KEY in config.py) require migration to enterprise secret management vaults before production cloud deployment.
- • **Synthetic Data Training:** ProductivAI's regression models were trained on mathematically simulated lifestyle data rather than longitudinal real-world student behavioral logs.
- • **Local SQLite File Locking:** Default local execution uses SQLite, which is prone to file-locking contention under high concurrent write loads compared to production PostgreSQL.

5.4 Future Scope

- • **WebSocket Real-Time Alerts:** Implementing Socket.IO WebSockets to replace HTTP polling, delivering instantaneous counsellor notification when emergency flags trigger.
- • **Longitudinal Trend Analysis:** Extending the AssessmentScore model to visualize student psychological trajectories across consecutive semesters using predictive time-series charts.
- • **Production Cloud Containerization:** Finalizing Dockerfile and Docker Compose configurations for seamless cloud deployment via Kubernetes or Render.
- • **Real-World Model Fine-Tuning:** Collecting opt-in, anonymized student academic performance and routine tracking data to fine-tune ProductivAI's predictive weights.

6. Conclusion

The AI Internship project successfully designed, implemented, and integrated two production-grade software platforms: the UAi PsychePortal clinical assessment suite and the ProductivAI lifestyle productivity engine. The project realized significant milestones across four engineering domains:

- **1. Full-Stack Web Engineering:** Engineered responsive, highly performant web applications combining Flask, SQLAlchemy, FastAPI, React 18, TailwindCSS, and Vite.
- **2. Clinical Data Systems:** Automated complex scoring mathematics, threshold triage, and dynamic multi-page ReportLab PDF generation for standardized psychological instruments.
- **3. Predictive Analytics:** Developed, trained, serialized, and deployed high-accuracy machine learning regression pipelines with sub-millisecond API response latency.
- **4. Architectural Integration:** Unified microservices, standalone desktop clients, and single-page apps into a cohesive, multi-process launchable ecosystem.

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PART 2: SUMMER INTERNSHIP WEEKLY REPORTS

Official Weekly Progress Reports (Week 1 through Week 6)
Submitted to: School of Ai & Future Technologies, Universal Ai University
Faculty Mentor: Prof. Rajesh Bhise | Department: Ai Cell

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Summer Internship Weekly Report

Student Details

Student Name: Himanshu Pathak	**Roll No.:** UAI05BT12417
Year/ Semester: 2nd Year / 4th Semester	**Email Id:** himanshu.pathak@universalai.in
Faculty Mentor : Prof. Rajesh Bhise	**Department:** School of Ai & Future Technologies

Company Details:

Company Name: Ai Cell, Universal Ai University	**Department Name:** School of Ai & Future Technologies
Company Manger Name: Prof. Rajesh Bhise / Dr. Shivaji Pawar	**Official Mail Id:**
Contact No:	

Project Title: UAi PsychePortal & AI Productivity Analyzer Suite
Week No.: First Week
Reporting Period: 4-May-2026 to 9-May-2026

1. Weekly Task

S. No	Task Description	Status
1	Requirement analysis, architectural design, and repository layout setup	Completed
2	Defined SQLAlchemy database schema (User, StudentProfile, Assessment)	Completed
3	Implemented Flask factory pattern and dynamic configuration handling	Completed

2. Technologies/Tools Used

- Python 3.13, Flask 3.0.0
- Flask-SQLAlchemy, SQLite
- Git, VS Code

3. Learning Outcomes

- Standardized database normalization techniques for relational models storing psychometric profiles.
- Configured Flask application factory patterns to isolate development and production environments.

4. Challenges Faced & Solutions

Challenges:
Avoid cyclic imports between models, database initializers, and authentication blueprints.

Solutions:
Implemented the Application Factory pattern (create_app) in app/__init__.py, decoupling model declarations from route handlers.

5. Plan for Next Week

S. No	Task to be done	Estimated Time
1	Implement DASS-21, BFI-44, EQ, and SQ mathematical scoring functions	6 Hours
2	Create automated risk-flagging logic for severe student scores	4 Hours
3	Develop seed data scripts (seed_questions.py, seed_demo_data.py)	4 Hours

Student Signature: Himanshu Pathak (Digital Signature - Verified)

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Year/ Semester: 2nd Year / 4th Semester	**Email Id:** himanshu.pathak@universalai.in
Faculty Mentor : Prof. Rajesh Bhise	**Department:** School of Ai & Future Technologies

Company Details:

Company Name: Ai Cell, Universal Ai University	**Department Name:** School of Ai & Future Technologies
Company Manger Name: Prof. Rajesh Bhise / Dr. Shivaji Pawar	**Official Mail Id:**
Contact No:	

Project Title: UAi PsychePortal & AI Productivity Analyzer Suite
Week No.: Second Week
Reporting Period: 11-May-2026 to 16-May-2026

1. Weekly Task

S. No	Task Description	Status
1	Implemented scoring algorithms for DASS-21, BFI-44, EQ, and SQ in app/scoring.py	Completed
2	Engineered automated emergency risk flagging for Moderate/Severe subscale categories	Completed
3	Created database population scripts for 100+ demo profiles and question banks	Completed

2. Technologies/Tools Used

- Python 3.13, SQLAlchemy ORM
- Flask-Bcrypt, Flask-Login
- SQLite database browser

3. Learning Outcomes

- Mastered reverse-score transformations (6 - R) and subscale matrix indexing for Big Five inventory metrics.
- Developed automated event triggers in SQLAlchemy to flag high-risk entries upon submission.

4. Challenges Faced & Solutions

Challenges:
Handling missing item selections without crashing subscale calculations.

Solutions:
Defaulted missing responses to neutral values (3) for BFI-44 and 0 for DASS-21 inside safe map retrievals (resp_map.get(i, default)).

5. Plan for Next Week

S. No	Task to be done	Estimated Time
1	Construct programmatic PDF report builder using ReportLab	6 Hours
2	Implement Excel data export using Pandas and OpenPyXL	4 Hours
3	Build Student Survey Web Portal (student-survey-portal/app.py)	5 Hours

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Company Manger Name: Prof. Rajesh Bhise / Dr. Shivaji Pawar	**Official Mail Id:**
Contact No:	

Project Title: UAi PsychePortal & AI Productivity Analyzer Suite
Week No.: Third Week
Reporting Period: 18-May-2026 to 23-May-2026

1. Weekly Task

S. No	Task Description	Status
1	Developed PDF export generation using ReportLab (SimpleDocTemplate, TableStyle)	Completed
2	Built Excel export features for counsellor metrics using Pandas and OpenPyXL	Completed
3	Deployed standalone Flask Student Survey Portal running on port 5001	Completed

2. Technologies/Tools Used

- ReportLab 4.1.0, Pandas 2.1.4
- OpenPyXL 3.1.2
- Flask Jinja2 Templating

3. Learning Outcomes

- Standardized programmatic document formatting using ReportLab flowables, tables, and page breaks.
- Integrated dual-Flask microservice instances sharing a common SQLite relational database.

4. Challenges Faced & Solutions

Challenges:
PDF layout text overflow when rendering student assessment details.

Solutions:
Encapsulated tabular cell text within Paragraph flowables with automated auto-wrapping text styles.

5. Plan for Next Week

S. No	Task to be done	Estimated Time
1	Develop modern React SPA Counsellor Dashboard using Vite and TailwindCSS	8 Hours
2	Build 11 dashboard view modules (Risk Alerts, Directory, Deployer)	5 Hours
3	Set up Chart.js radar and trend analytics visualizations	4 Hours

Student Signature: Himanshu Pathak (Digital Signature - Verified)

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Company Details:

Company Name: Ai Cell, Universal Ai University	**Department Name:** School of Ai & Future Technologies
Company Manger Name: Prof. Rajesh Bhise / Dr. Shivaji Pawar	**Official Mail Id:**
Contact No:	

Project Title: UAi PsychePortal & AI Productivity Analyzer Suite
Week No.: Fourth Week
Reporting Period: 25-May-2026 to 30-May-2026

1. Weekly Task

S. No	Task Description	Status
1	Built Vite + React single-page application inside uai-counsellor-dashboard/	Completed
2	Designed 11 dynamic views featuring TailwindCSS styling and Lucide icons	Completed
3	Configured interactive Chart.js visualizations for subscale trend analysis	Completed

2. Technologies/Tools Used

- React 18, Vite 5
- TailwindCSS 3.4, Lucide-React
- Chart.js 4.4

3. Learning Outcomes

- Engineered dynamic dashboard views using React state hooks (useState, useEffect).

- Applied responsive dark-theme design patterns using TailwindCSS utilities.

4. Challenges Faced & Solutions

Challenges:
Integrating client-side React routes into Flask server-rendered endpoints.

Solutions:
Compiled Vite static assets into Flask's static directory and rendered the root React via Jinja2 templates.

5. Plan for Next Week

S. No	Task to be done	Estimated Time
1	Inject Flask SQLite data directly into React global variables (window.__FLASK_KPIS__)	4 Hours
2	Develop ProductivAI synthetic dataset generator (50,000 samples)	4 Hours
3	Train and serialize Random Forest ML productivity regression model	5 Hours

Student Signature: Himanshu Pathak (Digital Signature - Verified)

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School of Ai & Future Technologies

Summer Internship Weekly Report

Student Details

Student Name: Himanshu Pathak	**Roll No.:** UAI05BT12417
Year/ Semester: 2nd Year / 4th Semester	**Email Id:** himanshu.pathak@universalai.in
Faculty Mentor : Prof. Rajesh Bhise	**Department:** School of Ai & Future Technologies

Company Details:

Company Name: Ai Cell, Universal Ai University	**Department Name:** School of Ai & Future Technologies
Company Manger Name: Prof. Rajesh Bhise / Dr. Shivaji Pawar	**Official Mail Id:**
Contact No:	

Project Title: UAi PsychePortal & AI Productivity Analyzer Suite
Week No.: Fifth Week
Reporting Period: 1-Jun-2026 to 6-Jun-2026

1. Weekly Task

S. No	Task Description	Status
1	Connected React frontend with Flask database using Jinja2 JSON script injection	Completed
2	Built 50,000-sample synthetic lifestyle dataset generator (generate_dataset.py)	Completed
3	Trained Random Forest Regressor and exported model via Joblib (productivity_model.pkl)	Completed

2. Technologies/Tools Used

- Scikit-Learn 1.3, Joblib
- NumPy, Pandas
- FastAPI, Uvicorn

3. Learning Outcomes

- Evaluated regression models using metrics such as R^2, MAE, and MSE.

- Deserialized trained .pkl estimators inside FastAPI endpoints for real-time inference.

4. Challenges Faced & Solutions

Challenges:
Model inference latency spikes when serving REST requests under heavy load.

Solutions:
Cached the deserialized Joblib model instance in global application memory upon FastAPI startup.

5. Plan for Next Week

S. No	Task to be done	Estimated Time
1	Develop ProductivAI React frontend featuring score dials and Recharts	5 Hours
2	Write integrated multi-process launcher scripts (launch_all.bat / .ps1)	4 Hours
3	Perform end-to-end system testing, bug fixing, and documentation writing	4 Hours

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Summer Internship Weekly Report

Student Details

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Year/ Semester: 2nd Year / 4th Semester	**Email Id:** himanshu.pathak@universalai.in
Faculty Mentor : Prof. Rajesh Bhise	**Department:** School of Ai & Future Technologies

Company Details:

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Company Manger Name: Prof. Rajesh Bhise / Dr. Shivaji Pawar	**Official Mail Id:**
Contact No:	

Project Title: UAi PsychePortal & AI Productivity Analyzer Suite
Week No.: Sixth Week
Reporting Period: 8-Jun-2026 to 13-Jun-2026

1. Weekly Task

S. No	Task Description	Status
1	Built ProductivAI React analytics dashboard with score dials and dark/light themes	Completed
2	Created unified multi-process orchestrators (launch_all.bat and launch_all.ps1)	Completed
3	Executed full system integration verification across ports 5000, 5001, and 8000	Completed

2. Technologies/Tools Used

- Windows Batch Scripting, PowerShell
- React 18, Recharts 2.10
- Framer Motion

3. Learning Outcomes

- Automated multi-process application deployment using shell scripting techniques.

- Verified cross-origin data flows and API request validation across distinct microservices.

4. Challenges Faced & Solutions

Challenges:
Managing multiple terminal windows and port collisions during full-stack integration.

Solutions:
Wrote automated PowerShell start routines with explicit port bindings (-port 5000, -port 5001, -port 8000).

5. Plan for Next Week

S. No	Task to be done	Estimated Time
1	Final project deliverable submission and demonstration to faculty mentors	0 Hours

Student Signature: Himanshu Pathak (Digital Signature - Verified)